Jira - Jira Automation Expansion

Jira automations areas, including scriptrunner, SIL, workflows, postfunctions, behavior, scripted CF,
Expand my list by covering all possible cases of automation

Below is a **comprehensive automation surface map for Jira (Data Center / Server, with Cloud notes where relevant)**. I've structured it by execution layer (event-driven, workflow-bound, UI-bound, data-bound, integration-bound, and platform-level) so you can reason about coverage, governance, and performance impact systematically.

You already listed:

- Automation
- ScriptRunner for Jira
- SIL (Simple Issue Language)
- Workflows
- Post-functions
- Behaviours
- Scripted Custom Fields

Below is the expanded, exhaustive view.

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1. Native Jira Automation Layers

1.1 Built-in Automation Engine

- Jira Automation (DC/Cloud)
  - Event triggers
  - Scheduled triggers
  - Webhook triggers
  - Manual triggers
  - Global/project scoped rules
  - Smart values / templating
  - Branching and parallel execution
  - Rate limits & execution logs

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2. Workflow-Level Automation

2.1 Workflow Constructs

- Status transitions
- Conditions
- Validators
- Post-functions
- Transition screens
- Global transitions
- Initial actions

2.2 Custom Workflow Extensions

Via:

- ScriptRunner for Jira
- Power Scripts for Jira (SIL engine)
- JMWE (Jira Misc Workflow Extensions)
- JSU Automation Suite

Automatable via:

- Custom Groovy conditions
- Scripted validators
- Scripted post-functions
- SIL-based transition logic
- Workflow listeners attached to transitions

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3. Event-Driven Automation (System-Level)

3.1 Jira Event Bus Listeners

- Issue created/updated/deleted
- Worklog events
- Comment events
- Attachment events
- Sprint events
- Project events
- User/group events

Implementation:

- ScriptRunner Listeners
- SIL Listeners
- Custom Java plugin listeners
- REST-triggered listener endpoints

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4. UI-Level Automation

4.1 Behaviours

- Dynamic field visibility
- Field requirement enforcement
- Field defaulting
- Cross-field logic
- Context-based UI manipulation

Via:

- ScriptRunner for Jira
- Power Scripts Behaviours (SIL)

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5. Field-Level Automation

5.1 Scripted Custom Fields

- Calculated fields
- Read-only computed values
- External REST-backed fields
- Derived SLA calculations

5.2 Database-backed Custom Fields

- Insight/Assets calculated fields
- SQL-backed fields (via plugins)
- External API fields

5.3 Field Configuration Automation

- Dynamic contexts
- Context switching via scripts
- Mass updates via REST

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6. Scheduled / Batch Automation

- Scheduled automation rules
- ScriptRunner Scheduled Jobs
- SIL Scheduled Tasks
- CRON-based jobs
- Database-level scheduled procedures (external)
- Jenkins-driven REST batch updates

- CLI automation (Atlassian CLI)

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7. REST/API-Level Automation

7.1 Internal REST

- Issue updates
- Bulk operations
- Transition triggers
- Workflow actions
- Filter manipulation
- Board configuration

7.2 External Integration Automation

- Webhooks (incoming/outgoing)
- Message queues (Kafka, RabbitMQ)
- Middleware (MuleSoft, Boomi)
- CI/CD integrations

Triggered from:

- Jira Software
- Bitbucket pipelines
- Jenkins
- GitHub Actions
- Azure DevOps

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8. Agile / Board-Level Automation

- Sprint creation automation
- Auto-assignment to sprint
- Sprint-complete transition logic
- Board quick filters logic
- Kanban WIP enforcement (via workflow + automation)

Applies to:

- Jira Software boards

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9. Assets / CMDB Automation

If using:

- Jira Service Management
- Assets (formerly Insight)

Automation includes:

- Object attribute updates
- Object-event triggers
- Object schema validators
- CMDB-based issue auto-population
- Auto-link issues to configuration items

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10. Service Management Automation

In:

- Jira Service Management

Includes:

- SLA breach triggers
- Queue-based automation
- Customer notification logic

- Auto-escalation
- Request-type routing
- Approval auto-processing
- Email channel parsing rules

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11. Security & Permission Automation

- Auto group membership via scripts
- Role assignment automation
- Permission scheme switching
- Issue Security Level automation
- Dynamic security level assignment
- Scripted user provisioning logic

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12. Marketplace App-Based Automation

Examples:

- ScriptRunner
- Power Scripts (SIL)
- JMWE
- JSU
- Automation for Jira (legacy plugin pre-native)
- Structure automation rules
- BigPicture event hooks
- Tempo triggers

Each app may:

- Add listeners
- Add workflow extensions
- Add custom REST endpoints
- Add background services

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13. Database-Level Automation (DC Only – High Risk)

Not recommended but seen in enterprise:

- Direct SQL triggers
- Scheduled DB jobs
- Materialized views for derived data
- Data repair procedures

Important in large DC instances like yours:

- Must respect entity_property, changegroup, changeitem integrity
- Avoid bypassing Lucene reindexing

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14. Cluster-Level Automation (Jira Data Center)

- Multi-node cache replication logic
- Hazelcast event propagation
- Node-local scheduled jobs
- Distributed locking for scripts
- Parallelized REST execution

In high-scale DC:

- ScriptRunner needs cluster-safe scheduling
- Avoid node-local state

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15. DevOps & CI Automation

- Auto-create issues from commits

- Smart commits
- Deployment gating via Jira status
- Auto-transition on PR merge
- Branch-based issue lifecycle control

Integrates with:

- Bitbucket
- GitHub
- GitLab
- Jenkins

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16. Email Automation

- Incoming mail handlers
- Comment-on-email reply parsing
- Auto-ticket creation
- Regex-based subject routing
- SIL mail handlers

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17. Notification & Communication Automation

- Custom notification schemes
- Scripted notifications
- Slack/MS Teams webhooks
- Conditional email suppression
- Smart recipient resolution

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18. Reporting / Derived Data Automation

- Scripted JQL functions
- Custom JQL extensions
- Calculated metrics fields
- Precomputed analytics
- REST-based data warehouse sync
- ELK/Splunk push jobs

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19. Migration & Bulk Automation

- CSV automation pipelines
- REST bulk import
- Project cloning scripts
- Workflow mass rewiring
- Bulk resolution fix scripts
- Data normalization scripts

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20. Governance & Compliance Automation

- Enforce resolution rules
- Enforce mandatory link types
- Prevent circular dependencies
- Validate JQL quality (relevant to your previous profiling work)
- Detect stale automation rules
- Audit rule execution frequency
- Automation drift detection

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21. Advanced Patterns (Enterprise DC)

Given your large-instance + profiling focus, these are critical:

21.1 Idempotent Automation Design

- Prevent re-trigger loops
- Use entity properties as markers
- Compare change history before updates

21.2 Parallel-safe Automation

- Distributed locks
- Optimistic concurrency control
- Retry-on-conflict patterns

21.3 Performance-aware Automation

- Avoid reindex storms
- Batch updates
- Async execution via REST
- Avoid heavy scripted CF recalculations

21.4 Automation Observability

- Rule execution telemetry
- Script execution timing logs
- Dead-letter queue patterns
- Centralized audit dashboards

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22. Custom Plugin-Level Automation (Java)

For ultimate control:

- Event listeners
- REST resources
- Servlet filters
- Scheduled services
- Indexing hooks
- Workflow module extensions
- Custom field types
- Custom JQL functions

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23. Edge Cases Often Missed

- Filter subscription automation
- Dashboard auto-provisioning
- Project template automation
- Portfolio/Advanced Roadmaps recalculation triggers
- Tempo timesheet enforcement
- Worklog validation scripts
- Cross-instance federation automation
- Smart value recursion traps
- Automation interacting with archived projects

Consolidated Automation Taxonomy

You can categorize every automation into:

1. **Trigger Source**
    - Event
    - Schedule
    - Manual
    - REST
    - UI
    - Database
    - External system

2. **Execution Layer**
    - Workflow
    - Listener
    - Scripted field
    - REST job
    - Plugin service
    - Cluster service
3. **Scope**
    - Global
    - Project
    - Board
    - Issue
    - Field
    - CMDB object
    - User
4. **Risk Profile**
    - UI only
    - Workflow blocking
    - Async background
    - DB-level
    - Cluster-wide

If you want, I can next provide:

- A **production-grade automation audit checklist for Jira Data Center**
- A **performance impact matrix (automation type vs scalability risk)**
- Or a **cluster-safe automation design framework for large Jira DC environments** aligned with multi-node parallel profiling.

Plan to find all existing in jira datacenter automations

Below is a **production-grade discovery and audit plan** to enumerate **all automation mechanisms in a large Jira Data Center instance**.
This is structured for **completeness, repeatability, and cluster safety**.

Scope assumption: enterprise-scale Jira DC (multi-node), multiple marketplace apps, custom scripts, and legacy artifacts.

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0. Define Automation (Audit Definition)

Before scanning, standardize what qualifies as automation:

Automation = *any deterministic logic that modifies state, enforces rules, or triggers actions without direct user intervention at execution time.*

Includes:

- Workflow logic
- Listeners
- Scheduled jobs
- Scripted fields
- UI dynamic logic
- REST-triggered logic
- Cluster services
- External systems triggering Jira
- Database-level jobs
- Mail handlers
- Notification hooks

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1. Instance-Wide App & Module Inventory

1.1 Enumerate Installed Apps

Export:

- UPM app list
- Version numbers
- Licensing state

Critical apps to inspect deeply:

- ScriptRunner for Jira
- Power Scripts for Jira
- Jira Service Management
- Jira Software

Reason: 80–90% of automation in DC lives in marketplace apps.

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2. Native Automation Engine Audit

2.1 Jira Automation Rules

Extract:

- Global rules
- Project rules
- Disabled rules
- Scheduled rules
- Webhook rules
- Rules with branch components
- Rules using advanced compare conditions

Collect:

- Trigger type
- Actor
- Last execution time
- Execution frequency

- Error rate

Risk markers:

- Rules updating issues in bulk
- Rules triggering transitions
- Rules modifying security level
- Rules without conditions
- Recursive patterns

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3. Workflow-Level Automation Discovery

3.1 Enumerate All Workflows

From DB:

- `jiraworkflows`
- `OS_WFENTRY`
- `OS_CURRENTSTEP`
- `OS_TRANSITIONINSTANCE`

Extract:

- Workflow name
- Active/inactive
- Project association
- Transition count

3.2 Per Workflow – Extract:

- Conditions
- Validators
- Post-functions
- Scripted components
- App-provided workflow modules

Flag:

- ScriptRunner script references
- SIL scripts
- Custom Java classes
- JMWE/JSU modules

Risk:

- Blocking validators with external REST calls
- Heavy scripted post-functions
- Hidden global transitions

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4. ScriptRunner Deep Discovery

If installed:

Enumerate:

4.1 Listeners

- Event type
- Script body
- Execution frequency

4.2 Scripted Fields

- Field name
- Template type
- Inline script vs file-based
- Cache usage
- External calls

4.3 Scheduled Jobs

- Cron expression
- Node execution mode
- Script logic

4.4 REST Endpoints

- Custom endpoints
- Authentication model
- External exposure

4.5 Behaviours

- Field dependencies
- Project scope
- Conditional logic complexity

4.6 Script Files (File System)

Scan:

```
<JIRA_HOME>/scripts/
```

Look for:

- Shared libraries
- Orphaned scripts
- Deprecated logic

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5. SIL / Power Scripts Discovery

If present:

Enumerate:

- SIL workflows
- SIL listeners
- SIL scheduled tasks
- SIL REST services
- SIL mail handlers
- Shared libraries

Look for:

- Global variables
- Cross-project script reuse
- External DB access

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6. Custom Field Automation Discovery

6.1 Identify Scripted/Calculated Fields

From DB:

- `customfield`
- `customfieldtypekey`

Look for:

- Scripted types
- Insight/Assets calculated fields
- Database-backed fields

For each:

- Evaluate evaluation mode (on view vs cached)
- Identify heavy search-based calculations

High-risk category in large DC.

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7. Listener Audit (Global)

From:

- App module descriptors
- Plugin XML
- System logs

Enumerate:

- All event listeners
- Event types subscribed
- Async vs sync execution

Critical events:

- Issue Updated
- Worklog Created
- Comment Created
- Sprint events

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8. Scheduled Job Discovery (Cluster-Level)

From:

- `clusteredjob`
- `scheduler_run_details`

Extract:

- Job class
- Node assignment
- Frequency
- Runtime
- Failures

Look for:

- Custom plugin schedulers
- ScriptRunner schedulers
- SIL schedulers
- Long-running jobs

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9. REST-Triggered Automation

Audit:

9.1 Incoming Webhooks

- Automation triggers
- ScriptRunner endpoints
- App endpoints

9.2 External Systems Calling Jira

Check:

- Access logs
- Reverse proxy logs
- API token usage

Look for:

- High-frequency issue updates
- Bulk transitions
- CI/CD triggers

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10. Mail Handlers

From:

- `mailserver`
- `mailhandler`

Enumerate:

- Handler class
- Scripted handler usage
- Regex routing rules
- Auto-transition logic

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11. Service Management Automation

If using JSM:
Audit:

- SLA escalation rules
- Approval auto-logic
- Request-type automation
- Assets object automation
- Queue-triggered rules

Heavy logic often hidden here.

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12. Agile / Board Automation

Audit:

- Sprint auto-close logic
- Auto-move to sprint rules
- Quick filter dynamic JQL
- Ranking automation
- Board-level triggers

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13. Database-Level Automation (Critical)

Inspect with DBA:

- SQL triggers
- Stored procedures
- Materialized views
- Cron-based DB jobs

Red flag:
Any automation modifying Jira tables directly.

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14. Notification Automation

Audit:

- Notification schemes
- Scripted notifications
- Custom email templates
- Slack/MS Teams integrations
- Webhook pushes

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15. Security Automation

Audit:

- Dynamic issue security level assignment
- Script-based role assignments
- Group auto-management scripts
- Project provisioning scripts

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16. Log-Based Discovery (Often Missed)

Scan logs for:

- Script execution markers
- Custom plugin class names
- REST endpoints invoked
- Scheduler executions
- Long-running script warnings

Pattern search:

```
ScriptRunner
SIL
Executing script
Listener invoked
```

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17. Orphan Detection

Look for:

- Disabled workflows still referenced
- Disabled automation rules
- Unused scripted fields
- Script files not referenced anywhere
- Jobs scheduled but doing nothing

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18. Performance Correlation Layer (Advanced)

Since you work with large DC environments:

Correlate automation with:

- Reindex spikes
- DB CPU spikes
- Cluster replication delays
- High `changegroup` growth
- Automation execution time logs

Build:

Automation Impact Matrix:
| Automation Type | Execution Mode | Sync/Async | Cluster-Wide | Risk |

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19. Output Deliverables

Produce:

1. Master Automation Inventory (spreadsheet)
    - Type
    - Owner
    - Scope
    - Risk rating
    - Performance impact
    - Last modified
    - Last executed
2. Orphaned Automation Report
3. High-Risk Automation Report
4. Cluster Safety Risk Report

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20. Final Validation Step

To ensure completeness:

Cross-check automation categories against:

- Event-driven
- Workflow-bound
- Scheduled

- Field-based
- UI-based
- REST-triggered
- DB-triggered
- External system triggered

If a category has zero entries — validate it truly doesn't exist.

If You Want Next

Given your DC scale and previous profiling work, I can generate:

- A **SQL-based discovery pack for Jira DC**
- A **REST-based automation crawler**
- A **cluster-safe parallel automation profiler**
- Or a **risk scoring model tailored for multi-node Jira DC**

Tell me your instance size (projects / issues / nodes) and I'll tailor it precisely.